Priority scheduling - SIVARATHINAM S (241401102)

Activities Terminal Apr 20 17:50

sivarathinam@localhost:~/sivarathinam/scheduling — /usr/bin/vim priority.c

```c
#include <stdio.h>
int main(){
	int n, completed=0, time=0;
	printf("Enter number of processes:\n");
	scanf("%d", &n);
	int at[n], bt[n], priority[n], done[n];
	int ct[n], tat[n], wt[n];
	float avgtat=0, avgwt=0;
	for(int i=0; i<n; i++){
		printf("Enter AT, BT and priority for P%d: ", i+1);
		scanf("%d %d %d", &at[i], &bt[i], &priority[i]);
		done[i]=0;
	}
	while(completed<n){
		int idx=-1;
		int highPriority=9999;
		for(int i=0; i<n; i++){
			if(at[i]<=time && done[i]==0){
				if(priority[i]<highPriorit){
					highPriority=priority[i
					[i];
					idx=i;
				} else if(priority[i]==highPriority){
					if(at[i]<at[idx])
						idx=i;
				}
			}
		}
		if(idx!=-1){
			time+=bt[idx];
			ct[idx]=time;
			tat[idx]=ct[idx]-at[idx];
			wt[idx]=tat[idx]-bt[idx];
			done[idx]=1;
			completed++;
		} else{
			time++;
		}
	}
	for(int i=0; i<n; i++){
		avgtat+=tat[i];
		avgwt+=wt[i];
	}
	printf("\nAverage Turnaround Time: %.2f\n", avgtat/n);
	printf("Average Waiting Time: %.2f\n", avgwt/n);
	return 0;
}
```

Output :

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sivarathinam@localhost:~/sivarathinam/scheduling — /usr/bin/vim priority.c

```
[sivarathinam@localhost scheduling]$ vi priority.c
[sivarathinam@localhost scheduling]$ gcc priority.c -o priority
[sivarathinam@localhost scheduling]$ ./priority
Enter number of processes:
4
Enter AT, BT and priority for P1: 0 4 2
Enter AT, BT and priority for P2: 0 3 1
Enter AT, BT and priority for P3: 0 2 3
Enter AT, BT and priority for P4: 0 1 4

Average Turnaround Time: 7.25
Average Waiting Time: 4.75
[sivarathinam@localhost scheduling]$
```